# CANTEEN HUB

Online Canteen Ordering, Payment & Digital Billing System

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| **Project Type** | Web-based canteen ordering and digital billing workflow |
| **Primary Users** | College students and canteen staff |
| **Core Idea** | Order online, pay first, receive a digital bill, then collect from the canteen |

## Project Documentation

Prepared as a project report. The proposed features and architecture can be updated to match the final implementation.

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# 1. Project Overview

## What is Canteen Hub?

Canteen Hub is a college-focused online canteen ordering system. Students can view available food items, place an order, and pay online before going to the canteen. Once payment is successfully verified, the order is confirmed and a digital bill is generated. The student then visits the canteen to collect the prepared order.

## Project summary

The system is designed to reduce time spent standing in long queues during busy college breaks. It does not provide delivery: the student’s responsibility is to collect the order from the canteen.

# 2. What Is Canteen Hub?

## Core concept

Canteen Hub connects a college canteen with students through a simple digital interface. Students browse the menu, select items and quantities, review the total, complete payment, and receive an order confirmation with a digital bill or receipt.

## Key rule

**An order is confirmed only after successful payment verification.** A pending, failed, cancelled, or unverified payment must not be treated as a confirmed order.

# 3. Problem Statement

## Current problem

College canteens often become crowded during lunch breaks and short intervals. Students may avoid buying food because the queue takes too long, or they may reach the counter only to discover that the item they want is unavailable.

## Problem to solve

Provide a convenient way to place and pay for an order before reaching the canteen, while keeping collection in person so the canteen does not need to offer delivery.

# 4. Why the Project Is Needed

## Student perspective

Students can place orders from their phone or computer, reduce time spent in the ordering queue, see the total before paying, and use a digital bill or order reference when collecting food.

## Canteen perspective

Staff can receive structured order details, including items, quantities, payment status, and order time. A digital bill also provides a clear record of what was ordered and paid for.

## College perspective

The project supports a more organized campus food service and can improve the student experience during short breaks.

# 5. Project Objectives

### Main objectives

• Reduce time spent waiting to place an order.
• Make the menu and prices easy to view.
• Allow students to choose items and quantities online.
• Require successful payment before order confirmation.
• Generate a digital bill showing the order and payment details.
• Provide an order ID or receipt for collection.
• Help canteen staff view and manage incoming orders.
• Keep the process simple: students collect orders themselves.

## 6. Scope and Users

### Student

Can browse the menu, add items to a cart, review the total, pay online, receive a digital bill, check order status, and collect the order. Account registration can be included if required.

### Canteen staff / administrator

Can manage menu items and availability, view paid orders, update preparation status, and mark an order as collected. Staff access should be protected.

### Included in scope

Menu browsing, cart, checkout, online payment, payment verification, order confirmation, digital bill, order status, and in-person pickup.

### Out of scope for the initial version

Home delivery, table service, and automatic delivery tracking. These are not required for the core project.

## 7. How the System Works

### Student workflow

1. Open Canteen Hub.
2. Browse the menu and check item availability.
3. Select food items and quantities.
4. Review the cart, item prices, and total amount.
5. Proceed to online payment.
6. The system verifies the payment result.
7. If payment succeeds, the system confirms the order, assigns an order ID, and generates a digital bill/receipt.
8. The student saves or shows the digital bill/order ID at the canteen.
9. Canteen staff prepare the order, verify the order details, hand over the food, and mark it as collected.

### Payment failure

If payment fails, is cancelled, or cannot be verified, the system must not show the order as confirmed and must not issue a paid receipt. It can show a pending-payment message and let the student retry or check status.

### Important implementation note

The backend should verify payment using the payment provider’s trusted response or webhook. It should not rely only on a student returning to a success page or on a value sent by the browser.

## 8. Functional Requirements

### FR-1: Menu

Display food names, prices, and availability.

### FR-2: Cart

Allow students to add, remove, and change item quantities before payment.

### FR-3: Total calculation

Calculate the total using trusted item prices from the backend.

### FR-4: Payment

Initiate payment through a supported payment gateway or approved payment method.

### FR-5: Payment verification

Verify the payment result before confirming the order.

### FR-6: Order confirmation

After successful payment, generate a unique order ID and confirmation details.

### FR-7: Digital bill

Generate an electronic bill/receipt for a confirmed, paid order.

### FR-8: Order management

Allow authorized canteen staff to view paid orders and update their status.

### FR-9: Pickup

Let the student show the order ID or digital bill; staff mark the order as collected.

### FR-10: Availability

Prevent checkout for unavailable items and recheck availability before payment.

## 9. Non-Functional Requirements

### Usability

The interface should be simple and mobile-friendly, with readable menu items, clear totals, and understandable payment messages.

### Performance

Menu pages and order status should load promptly under normal campus usage.

### Security

Use HTTPS in deployment, validate inputs on the server, protect staff/admin routes, and never store raw card details.

### Reliability

Prevent duplicate confirmations or duplicate bills when a payment notification is received more than once.

### Privacy

Collect only the student information needed to operate the service and protect it from unauthorized access.

### Maintainability

Keep menu, payment, billing, and order-management logic organized so each can be updated independently.

# 10. Proposed Modules and Features

### Student module

Menu, cart, checkout, payment, digital bill, confirmation, and order-status view.

### Authentication module (if used)

Student sign-in and protected staff/admin sign-in.

### Menu management module

Create or update food items, prices, and availability.

### Payment module

Create payment requests, verify provider responses, and handle failed or pending payments.

### Digital billing module

Create a bill after payment is verified. Show the order number, date/time, item names, quantities, per-item prices, subtotal, applicable taxes/charges if any, total paid, payment status, and payment reference where available.

### Order management module

Show confirmed paid orders to staff, update preparation status, and mark collection.

### Admin dashboard

A single place for authorized staff to manage menu items, orders, and order status.

# 11. Digital Bill / E-Receipt

### Purpose

The digital bill gives the student proof of the items ordered and payment completed, without needing a paper bill. It also helps staff match the student's pickup to the correct order.

### Suggested bill fields

- Canteen name and Canteen Hub branding.
- Unique order/bill number.
- Order date and time.
- Item name, quantity, unit price, and line total.
- Subtotal and any applicable tax/extra charge.
- Final amount paid.
- Payment status and transaction reference, if available.
- Pickup status or current order status.

### When it is generated

Generate the paid receipt only after the backend has verified successful payment. If payment is pending or failed, show payment status separately and do not label it as a paid bill.

### How the student uses it

The student can view the bill on the confirmation screen and, if implemented, download it or access it later from order history. At pickup, the student shows the bill or order number to canteen staff.

### Important distinction

A digital bill is the receipt for the order/payment. It does not mean food is delivered; the student still collects the order from the canteen.

# 12. Data and Order Status

### Suggested records

Student: account ID and required details. Menu item: item ID, name, price, availability. Order: order ID, student reference, item snapshot, quantities, total, payment reference, timestamps, and status. Bill: bill number, order ID, item breakdown, total paid, and issue time.

### Suggested order statuses

Awaiting Payment → Confirmed / Paid → Preparing → Ready for Pickup → Collected. A payment that fails or is cancelled should be marked Failed or Cancelled, not Confirmed.

### Payment status versus order status

Keep payment status separate from food-preparation status. For example, payment can be Paid while the order is Preparing. This makes the system easier to audit and troubleshoot.

# 13. Suggested System Architecture

### Frontend

A responsive web interface for viewing the menu, managing the cart, paying, and viewing the digital bill and order status.

### Backend

An API that validates items and prices, creates orders, communicates with the payment provider, verifies payment, generates the bill, and serves authorized staff actions.

### Database

Stores menu items, order details, payment references, digital bill details, and status history.

### Payment provider

A supported payment gateway handles payment securely. The backend verifies payment before confirming an order and issuing a paid receipt.

### Technology note

The exact framework, programming language, database, and payment gateway should be documented once the project stack is finalized. This report intentionally describes a technology-neutral design.

# 14. Benefits

### For students

Less time spent in the ordering queue, convenient menu browsing, clear totals, a digital receipt, and a confirmation reference for pickup.

### For canteen staff

Structured paid orders, clearer pickup verification, and a digital record of what was ordered.

### For the college

A practical digital solution to a common campus problem that can be expanded after the basic workflow works reliably.

# 15. Limitations and Risks

### Internet or payment issues

Students may have weak connectivity or a payment may remain pending. Show clear status messages and provide a safe retry/check-status option.

### Peak demand

Online ordering can move the queue from the counter to food preparation. The canteen may need preparation-time estimates, order limits, or scheduled pickup slots.

### Item availability

An item may sell out after it is added to the cart. Recheck availability before payment and explain any change to the student.

### Refunds and cancellations

Define a clear policy for failed payments, duplicate payments, unavailable items, and cancellations before launching.

### Payment security

Use a reputable payment provider and never treat a client-side success message alone as proof of payment.

## 16. Future Enhancements

### Possible improvements

- Downloadable PDF bill or printable receipt.
- Order history and bill re-download.
- Estimated preparation time and pickup time slots.
- Notifications when an order is ready.
- Daily menu and sold-out indicators.
- Student feedback and ratings.
- Sales and popular-item reports for canteen staff.

## 17. Testing Plan

### Core test cases

1. Menu loads and displays correct prices.
2. Student can add and remove cart items.
3. Total updates correctly when quantities change.
4. Successful payment leads to one confirmed order and one digital bill.
5. Failed or cancelled payment does not confirm an order or issue a paid receipt.
6. Repeated payment notifications do not create duplicate orders or bills.
7. Unavailable items cannot be checked out.
8. Unauthorized users cannot access staff functions.
9. Staff can update an order to Ready for Pickup and Collected.
10. Student and staff can match the digital bill/order ID to the correct order.

### Acceptance criteria

The project is ready for a basic demonstration when a student selects items, pays successfully in a test environment, receives a confirmed order ID and digital bill, and collects the matching order, while failed payments remain unconfirmed.

## 18. Conclusion

## Summary

Canteen Hub is designed to make college canteen ordering more convenient during crowded breaks. Students place orders online and pay before the order is confirmed. After payment verification, the system generates a digital bill and sends the order to the canteen for preparation. The student collects it in person. The menu, payment, confirmation, billing, and pickup workflow addresses the main problem without adding delivery complexity.